

Rayan Mohammed

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Profile

Software engineer (B.Tech CS, VIT 2026) who builds and ships full-stack features end to end. Designed a hybrid DQN + NSGA-II scheduling algorithm that beat six optimization baselines by 38–44% ($p < 0.001$), and builds backend REST systems and automation pipelines (Django, Docker, n8n). Strong in data structures, algorithms, distributed systems, and problem-solving.

Skills

Languages	Python, Java, JavaScript, SQL
Backend	Django, REST API Development, MySQL
Data & ML	PyTorch, Pandas, NumPy, Scikit-learn
Tools & Cloud	Git, Docker, Docker Compose, n8n, Linux, AWS (EC2, S3, IAM, VPC, Auto Scaling, ELB)
Core	Data Structures, Algorithms, OOP, DBMS, Distributed Systems

Experience

CVS Info Solutions, Infopark July 2024

Software Development Intern

- Debugged REST API endpoints and backend modules to resolve authentication, data-flow, and user-access defects across the application stack
- Implemented and analyzed role-based access control (RBAC) to secure user and admin workflows
- Investigated API failures through log analysis to improve database query performance, and documented recurring issues to streamline debugging cycles

MicroIT – Student Tech Initiative December 2023 – January 2025

Co-Founder

- Shipped core backend features for a live B2B travel platform (VayoAura) serving real customers, owning development end to end in a startup-pace environment
- Led API design and system architecture for web and mobile, translating client requirements into deployed backend workflows from spec to production

AIESEC in Amaravati August 2024 – May 2025

Business Development Associate

- Ran outbound prospecting to 40–60 B2B leads per week across email, LinkedIn, and cold calls, booking 5–6 discovery calls and partner meetings per month
- Qualified leads by fit and intent and maintained a structured pipeline with disciplined multi-touch follow-ups

Projects

Task Scheduling Optimization in Distributed Systems *Python, PyTorch, Alibaba Cluster Trace*

- Designed a hybrid DQN + NSGA-II + Honey Badger algorithm for multi-objective scheduling on real Alibaba traces, outperforming GA, PSO, GWO, EHO, HBO, and RL baselines by 38–44% ($p < 0.001$, Wilcoxon rank-sum, 30 runs)
- Built a Deep Q-Network manager that adaptively selects among 7 evolutionary operators via experience replay and ϵ -greedy exploration, reaching HEFT-optimal makespan on a 450-task, 10-processor workload
- Engineered a DAG-based scheduling simulator with HEFT ranking, SLA deadline enforcement, and joint optimization of makespan, energy, wait time, and utilization; validated across ZDT and DTLZ benchmark suites

Clara AI Automation Pipeline *Docker, Docker Compose, n8n*

- Built a containerized pipeline that turns AI call transcripts into structured agent-configuration artifacts, using version-controlled n8n workflows for data extraction and JSON memo generation
- Improved reproducibility and debugging through isolated service deployment and modular orchestration of transcript ingestion, transformation, and artifact generation

VSync – Smart College Companion Platform *Flutter, Dart, REST API*

- Built a modular full-stack academic-management app with a Flutter front-end and REST API integration, including GPA computation from structured academic records (published on Google Play)
- Implemented reusable UI components and dynamic state management using clean-architecture principles

Gesture-Controlled Drone *Raspberry Pi, OpenCV, Python*

- Developed a real-time computer vision system mapping hand gestures to drone flight controls, with OpenCV pipelines optimized for low-latency response

Education

Vellore Institute of Technology (VIT) 2022 – 2026

B.Tech in Computer Science Engineering, Graduated 2026

Certifications

AWS Academy Cloud Architecting (2024) | AWS Academy Cloud Foundations (2024) | CouchDB Distributed Database System (2024)